MIL-P-45661 (Ord)
26 February 1960
SUPERSEDING
FA(FC)-PD-39
21 February 1958

MILITARY SPECIFICATION

PERISCOPE, TANK: M27

1. SCOPE

1.1 This specification covers one type of periscope used as a vision device in combat type vehicles.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

L-P-406	Plastics, Organic; General Specifications, Test Methods

Military

MIL-P-11901	Periscopes M6, M12, M13, M13B1, M14, M14A1, M17, M26, and M27; Packaging of
MIL-O-13830	Optical Components for Fire Control Instruments; General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of

FSC 6650

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:41Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U. S. Military Property

DRAWINGS

Ordnance Corps

F7633132	Periscope, Tank: M27
IEL 7633132	Inspection Equipment List for Periscope, Tank: M27

(Copies of specifications, standards, and drawings required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Production model. - At the beginning of production one periscope, representative of production, shall be made available to the Government to determine the ability of the periscope to meet all requirements of this specification.

3.2 Materials. - Materials shall be in accordance with the drawings, general specifications, and material specifications forming a part of this specification.

3.3 Construction. - The periscope shall be manufactured in accordance with Drawing F7633132, "Periscope, Tank: M27", and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to obtain the desired mechanical characteristics, performance and permanence to insure inherent quality and performance capabilities prior to final acceptance by the Government.

3.4 General specifications. - The periscope shall meet the following requirements of Specifications MIL-F-13926 and MIL-O-13830.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:41Z
Check the source to verify that this is the current version before use.

3.4.1 MIL-F-13926.

(a) Order of precedence
(b) Protective finished surface
(c) Special accuracy
(d) Finish, painting and protective
(e) Workmanship

3.4.2 MIL-O-13830.

(a) Cleanliness

3.5 Environmental.

3.5.1 Shock.- The periscope shall not be damaged after being subjected to the shock test specified in Specification MIL-O-13830 after which it shall meet all requirements of the specification.

3.5.2 Temperature.- The periscope shall show no evidence of physical failure and shall meet the requirements of this specification at room temperature (60 to 90 degrees Fahrenheit) after being exposed to ambient temperatures of plus 160 and minus 65 degrees Fahrenheit.

3.5.3 Mirror and window laminations.- Mirror and window laminations shall not be damaged when subjected to the temperatures specified in 3.5.2. The appearance of bubbles, blisters or other indications of cement separation shall be considered evidence of bond failure.

3.6 Optical characteristics.

3.6.1 Definition.- The periscope shall permit resolution of test pattern subtending one minute of arc within the resolving area as outlined on Drawing F7633132.

3.6.2 Spherical power.- Spherical power shall be within the range of minus 0.50 to plus 0.25 diopter.

3.6.3 Astigmatism.- Astigmatism shall not exceed 0.25 diopter.

3.6.4 Light transmission.- Light transmission shall not be less than 50 percent.

3.7 Plastic body material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:41Z
Check the source to verify that this is the current version before use.

3.7.1 Heat distortion temperature. - Heat distortion temperature shall be between 150 and 190 degrees Fahrenheit when measured in accordance with Specification L-P-406, Method 2011.

3.7.2 Hardness. - The Rockwell number of the plastic body shall be not less than M80 when measured in accordance with Specification L-P-406, Method 1081.

3.7.3 Color. - Color of the body shall be crystal clear throughout, as defined by the standard furnished by the contracting officer ~~shown~~ Showing lower limit of acceptability.

3.7.4 Bubbles and inclusions (excepting lint). - Bubbles and inclusions (excepting lint) shall not exceed 0.040 inch in maximum diameter and the total projected area (perpendicular to the light path) of all bubbles and inclusions shall not exceed 0.05 percent of the cross-sectional area of the plastic body perpendicular to the light path. There shall be not more than five 0.040 inch, or ten 0.030 inch, or twenty-one 0.020 inch, or eighty-three 0.010 inch, or three hundred and thirty-two 0.005 inch diameter inclusions (excepting lint) or any combination of bubbles and inclusions of these sizes totaling to more than the allowed projected cross-sectional area. No lint particles shall exceed 0.025 inch in projected length (perpendicular to the light path). Not more than one such lint particle shall be permitted. No additional lint particles shall exceed 0.125 inch in projected length. The cross sectional area of all lint particles shall be computed on the basis of 0.003 inch width and included in the computation of total projected area. The total projected length of all lint particles shall not exceed 1.35 inches. The maximum permissible combination of lint with inclusions shall not be permitted.

3.8 Identification marking. - Identification marking, unless otherwise specified, shall be in accordance with Standard MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor inspection. - Unless otherwise specified herein, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:41Z
Check the source to verify that this is the current version before use.

4.1.1 Contractor's quality assurance system. - The contractor shall provide and maintain an effective quality assurance system acceptable to the Government covering the supplies under this contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when it provides:

(a) The quality assurance required by the applicable drawings and specifications listed in the contract or order.

(b) Contractor's quality standards of acceptability.

(c) Procedure for continually evaluating product quality.

(d) Procedure for continaully reporting evidence of actual product quality.

(e) Measures for preventing any significant departure of product quality from standards.

4.1.2 The contractor will not be restricted to the inspection station or to the method of inspection listed, provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether or not certain procedures of the system provide equal assurance, the comparable procedure of this specification shall be used. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification or other documents referenced herein.

4.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at unscheduled intervals. Verification will consist of (a) surveillance of the operations to determine whether the practices, methods, and procedures of the written inspection plan are being properly applied, and (b) Government product inspection to measure quality of product offered for acceptance. Deviation from the prescribed or agreed-upon procedures, or instances of poor practices which might have an affect upon the quality of the product, will immediately be called to the attention of the contractor. Failure of the contractor to promptly correct deficiencies discovered shall be cause for suspension of acceptance until correction has been made or until conformance of product to prescribed criteria has been demonstrated. To avoid interference with operations, the contractor will designate a responsible official or officials to whom the Government inspector will report such instances.

4.3 Production model. - Unless otherwise specified in the contract, one periscope taken from the production run shall be tested by the contractor under surveillance of the Government. These tests shall be conducted at the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:41Z
Check the source to verify that this is the current version before use.

contractor's facilities or Government approved laboratory using test equipment designed or procured for use on the remaining items. The model shall meet all the requirements of this specification using the inspection methods prescribed herein.

4.3.1 Disposition of rejected production model. - Should the production model fail to pass the test for Government acceptance, the contractor shall cease those operations which are causing rejectable material and make the necessary corrections. When corrections have been made, a new production model shall be tested by the contractor under surveillance of the Government. Operations causing rejections shall not be resumed until a production model has successfully passed all inspections specified.

4.4 Inspection provisions.

4.4.1 Lot formation. - Unless otherwise specified by the contracting officer, or his representative, inspection lot sizes and lot formation shall be in accordance with Standard MIL-STD-105.

4.4.2 Examination. - The classification of defects in Table I shall constitute the minimum inspection requirements to be performed by the contractor prior to acceptance or rejection by lot. The Government reserves the right to inspect for any applicable requirement, and to reject individual non-conforming items.

4.4.3 Disposition of rejected lots. - Rejected lots shall be screened for all defective characteristics. After corrections have been made and the Government inspector informed of the corrective action taken, the lot shall be resubmitted for acceptance as outlined in Standard MIL-STD-105.

4.4.4 Sampling. - Samples for tests listed in Table I shall include the samples that have successfully passed the temperature requirements of 4.7.1.2. Temperature sampling shall use Level L5, AQL 4.0 percent, and Sampling Plan Table IV-A of Standard MIL-STD-105.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:41Z
Check the source to verify that this is the current version before use.

TABLE I - CLASSIFICATION OF DEFECTS

Use Inspection Level II, and Sampling Plan Table IV-A of Standard MIL-STD-105

MAJOR: AQL 1.5%	Req't Par.	QA Par.	Test Method
1. Definition	3.6.1	4.7.2.1	Device, Special Testing (DST)
2. Spherical power	3.6.2	4.7.2.2	DST
3. Astigmatism	3.6.3	4.7.2.3	DST
4. Light transmission	3.6.4	4.7.2.4	Standard Measuring Equipment (SME)
5. Heat distortion	3.7.1	4.7.3.1	DST
6. Hardness	3.7.2	4.7.3.2	SME
7. Color	3.7.3	4.7.3.3	Visual
8. Bubbles and inclusions	3.7.4	4.7.3.4	SME
9. Identification marking	3.8	4.8	Visual

NOTE: These tests shall be conducted at standard ambient temperature (+60°F to +90°F)

4.5 Inspection testing.

4.5.1 Control of materials. - Materials shall be tested to determine conformance to contractually applicable specifications. Such tests may be conducted in the prime contractor's or subcontractor's facilities, or in other suitably equipped laboratories. The test results shall be available to the Government inspector for review. Certified test reports, identifiable with the material, may be accepted in lieu of repeated tests. The Government reserves the right, in any case, to require retests to assure that the materials are acceptable.

4.6 Inspection equipment.

4.6.1 Contractor provisions. - Unless otherwise specified in the contract, the contractor shall provide and properly utilize all gaging, measuring, and testing equipment of the required accuracy for the performance of all inspection examinations and tests required to be performed by the contractor under this contract. The contractor shall be responsible for the maintenance of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:41Z
Check the source to verify that this is the current version before use.

equipment including surveillance, calibration, and rehabilitation, on a regularly scheduled basis, and shall have available, all calibration equipment and other instrumentation essential for calibration and certification of the inspection equipment. The contractor also shall maintain records of inspection equipment surveillance and calibration, and shall make these records available for review by the Government. The Government reserves the right to certify inspection equipment, and to use it for Government verification inspection or acceptance inspection.

4.6.2 Accuracy requirements. - Inspection equipment which incorporates features for measuring values or sizes, established by product requirements, shall be capable of measuring these values within an accuracy of ten percent of the specified tolerance. In cases where the ten percent accuracy imposes impractical or very difficult design and fabrication problems, or results in a test equipment cost factor which is exorbitant in relation to the cost and application of the product to be inspected, the Government shall be expeditiously advised in order that necessary appropriate action can be taken.

4.6.3 Device to test definition, spherical power and astigmatism. - Test Fixture F8288800 shall be utilized for Government verification. All test equipment utilized for acceptance inspection of functional requirements shall be correlated with Test Fixture F8288800, as shown on IEL-7633132.

4.6.4 Device to test shock requirements. - A special testing device, the accuracy of which, shall be in strict accordance with all specified accuracy requirements contained in, and pertinent to, Specification MIL-O-13830.

4.6.5 Government provisions. - Test equipment listed in Inspection Equipment List, IEL 7633132 for Periscope, Tank: M27, will be used for Government production inspection (reference 4.2 (b)), and to settle disputes (reference 4.1.2). Drawings of this equipment may be furnished the contractor for informational purposes.

4.7 Test methods and procedures.

4.7.1 Environmental.

4.7.1.1 Shock test. - The periscope shall be mounted in its normal operating position and shocked in a direction parallel to the vertical line of sight of the periscope. After shock, the periscope shall be visually examined for damage. It shall then be subjected to all the test requirements of this specification. This test shall be performed on the production model only.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:41Z
Check the source to verify that this is the current version before use.

4.7.1.2 Temperature. - The periscope shall be placed in a carton insulated with one inch of fiberglass. The open carton shall be exposed to the ambient temperature of minus 65 degrees Fahrenheit until thermal stabilization is reached. The carton shall then be sealed and placed at room temperature for twenty-four hours; upon completion of this period, the carton shall be opened and the periscope examined visually for physical defects. Tests for conformance with 3.5.2 shall be conducted not less than one week later, after which the instrument shall then be subjected to plus 160 degrees Fahrenheit until thermal stabilization is reached. The periscope shall be returned to room temperature within a ten hour period and after 24 hours at room temperature, the periscope shall be inspected for conformance with 3.5.2.

4.7.1.3 Mirror and window laminations. - The laminations of mirror and window shall be visually examined after exposure to the temperature test as specified in 4.7.1.2. The appearance of bubbles, blisters, or any cement separation shall be considered evidence of bond failure.

4.7.2 Optical tests.

4.7.2.1 Definition test. - A test target shall be used, the pattern of which is illustrated in Specification MIL-O-13830, Figure 1. The image shall be viewed with the aid of a three-power auxiliary telescope scaled in diopters with an objective aperature of one-half inch. The test pattern shall be resolved in each of the four meridians by an adjustment of the auxiliary telescope not to exceed plus or minus 1/8 diopter from the setting for best focus on the numeral in the center of the test pattern. Illumination of the test chart shall be a minimum of ten millilamberts. At least three regions of the clear aperture of the periscope, mutually separated by 1-1/2 inches or more, shall be inspected for definition.

4.7.2.2 Spherical power test. - Using the test target and auxiliary telescope as specified in 4.7.2.1, the spherical power, when checked in at least three regions of the clear aperature of the periscope, mutually separated by 1-1/2 inches or more, shall be within the specified range.

4.7.2.3 Astigmatism test. - Astigmatism shall be tested using the test target and auxiliary telescope as specified in 4.7.2.1, and shall be tested in at least three regions of the clear aperature of the periscope, mutually separated by 1-1/2 inches or more. It shall not exceed the specified diopter reading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:41Z
Check the source to verify that this is the current version before use.

4.7.2.4 Light transmission test. - This test may be performed in conjunction with the "bubbles and inclusions test" specified in 4.7.3.4 using standard measuring equipment. It shall not exceed the specified requirements for light transmission.

4.7.3 Plastic body material.

4.7.3.1 Heat distortion test. - The heat distortion temperature test shall be in accordance with Specification L-P-406, Method 2011.

4.7.3.2 Hardness test. - The hardness of the plastic body shall be tested in accordance with Specification L-P-406, Method 1081, using standard measuring equipment. The Rockwell reading shall not be less than specified.

4.7.3.3 Color test. - This test shall be visual and may be conducted in conjunction with the bubbles and inclusions test. The color shall not exceed the lower limit of Standard Color Comparison B7681038, shown on IEL 7633132, as specified by the contracting officer.

4.7.3.4 Bubbles and inclusions test. - The samples of plastic material for the body shall be inspected for bubbles, inclusions and lint under brilliant illumination such as is provided with a standard 500 watt lantern slide projector, or equivalent, placed at one end of the blank. The viewing shall be through one of the faces of the blank, to which a heavy coat of mineral oil has been applied to produce a transparent surface. The blank shall be backed by a dull background such as black felt or black masking paper during this inspection.

4.8 Identification marking. - Marking shall be visually inspected for conformance to Standard MIL-STD-130.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Specification MIL-P-11901.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Selections of applicable levels of preservation, packaging, and packing
(c) Applicable stock number.
(c) Provisions for production model including date of delivery.

Source: https://assist.dla.mil - Downloaded: 2026-10-22T03:41Z
Check the source to verify that this is the current version before use.

6.2 Definitions.

6.2.1 Standard measuring equipment (SME). - Standard measuring equipment (SME) includes the common hand-type measuring devices which are usually stocked by commercial supply houses for ready supply (shelf items), and which are normally used by an inspector to perform dimensional inspection of items under procurement. This category also includes commercial testing equipment such as meters, optical comparators, etc.

6.2.2 Device, special testing (DST). - Special testing devices are defined as special inspection fixtures, instruments, and modified standard measuring equipment.

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any invention that may in any way be related thereto.

Custodian:

Army - Ordnance Corps

Preparing Activity:

Army - Ordnance Corps

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T03:41Z
Check the source to verify that this is the current version before use.